**Unit:** 6 Unit 5: Linear Equations
**Topic:** Rate of Change and Slope

**Name:** ____________________
**Date:** ______________

## Advanced (Carolina Reaper)

**Q1.** What is the slope of the line that passes through points $(2, 3)$ and $(4, 5)$?
(A) $\frac{1}{2}$
(B) 1
(C) $\frac{3}{2}$
(D) 2
(E) $\frac{5}{2}$

**Q2.** A water tank is being filled at a rate of 2 gallons per minute. What is the slope of the graph of the amount of water in the tank versus time?
(A) 0
(B) 1
(C) 2
(D) $\frac{1}{2}$
(E) $-2$

**Q3.** The cost of renting a bike is $10plus$ 2 per hour. What is the slope of the graph of the cost versus time?
(A) 2
(B) 5
(C) 10
(D) $\frac{1}{2}$
(E) 0

**Q4.** A bookshelf has 5 shelves, and each shelf can hold 8 books. If the bookshelf is currently empty, what is the slope of the graph of the number of books on the bookshelf versus the number of hours spent stocking the bookshelf at a rate of 2 books per hour?
(A) $\frac{1}{4}$
(B) 1
(C) 2
(D) 4
(E) 8

**Q5.** A person is saving money for a trip and has $100alreadysaved.Theyplantosaveanadditional$ 20 per week. What is the slope of the graph of the amount saved versus time?
(A) 10
(B) 20
(C) 50
(D) 100
(E) 200

**Q6.** What is the slope of the line $y = -3x + 2$?
(A) $-1$
(B) $-2$
(C) $-3$
(D) 2
(E) 3

**Q7.** A person is planning a road trip and wants to drive 250 miles in 5 hours. What is the slope of the graph of the distance traveled versus time?
(A) 25
(B) 50
(C) 50
(D) 60
(E) 100

**Q8.** What is the slope of the line that passes through points $(1, 2)$ and $(3, 2)$?
(A) 0
(B) 1
(C) $\frac{1}{2}$
(D) 2
(E) undefined

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**Open Ended Questions (FRQ)**

**Q9.** A bakery is making cakes for a wedding. They need to make 150 cakes in 10 hours. If they have already made 20 cakes in 2 hours, what is the slope of the graph of the number of cakes made versus time? Show your work and explain your answer.

**Q10.** A person is saving money for a down payment on a house and has $500alreadysaved.Theyplantosaveanadditional$ 50 per month. If they want to have $2000savedin2years, whatistheslopeofthegraphoftheamountsavedversustime?Showyourworkandexplainyouran$

**Chef's Tips**

- Tip 1: Make sure to read each question carefully and understand what is being asked.
- Tip 2: Use the given information to find the slope, and show your work.
- Tip 3: Check your answers to make sure they make sense in the context of the problem.